OPERATION & MAINTENANCE MANUAL
FOR
ELECTRO HYDRAULIC DOCK LEVELER

CLIENT: ZAMIL STEEL BUILDINGS INDIA PVT LTD.
MFG.YEAR: - FEB 2018.

DESIGNED, DEVELOPED & MARKETED
BY,

NIHVA

NIHVA Technologies Pvt. Ltd.

G-BLOCK, 1ST FLOOR, THERMAX CHOWK
MIDC, ABOVE RENAULT SHOWROOM
CHINCHWAD, PUNE-411019

TEL.NO: - +912066304033
EMAIL: -info@nihva.com
www.nihva.com

A. DESPATCH & UNLOADING

1. Loading, for dispatch, will be our responsibility. On receipt of the Dock-leveler at site, please check and ensure that there is no physical damage to the equipment.

2. While unloading the equipment from the transporting vehicle, please ensure to use slings with rated capacity higher than the gross weight of the Stationary lift in packed condition.

3. Slings MUST be hooked ONLY at specified positions. These positions are indicated and marked with a figure of 'HOOK'. Improper sling position or improper slinging could cause damage to the equipment.

B. ORDERING OF SPARES

1. When ordering for spare, please specify the following in your Purchase Order.

    - Equipment Name
    - Model Number
    - Equipment Serial No.
    - Part name / Description
    - Part Number
    - Quantity required
    - Delivery period / time
    - Mode of dispatch

    This would avoid ambiguity and help us serve you better.

DATASHEET FOR DOCK LEVELER

CLIENT:

1.0 CONSTRUCTION DETAILS

1.1 Required Capacity - each	: 6 ton
1.2 Model No.	: TEDL-2000
1.3 Leveler operation mode operated	: AC power pack hydraulically
1.4 Lip operation mode	: hydraulic
1.5 Operation control	: by push buttons and control panel
1.6 Loading platform height from road Level	: 1200 mm (Will be confirmed during Detailing)
1.7 Travel above & below dock level	: (+) 350 / (-) 250 mm
1.8 Platform Size (L x W)	: 2940 x 2070 mm
1.9 Lip size (L x W)	: 2070 x 400 mm
1.10 Maximum slope can be achieved	: 8 to 10%
1.13 Clear Pit dimensions (L x W x D)	: 2990x 2120 x 485 mm
1.14 Hydraulic power pack	
1.14.1 Make	"NIHVA"
1.14.2 Driver type	: Direct by electric Motor
1.14.3 Absorbed power	: 0.75 kW
1.14.4 Motor Ratings	: 0.75/1 kW /HP
1.14.5 Degree of protection for motor	: IP 55
1.14.6 Class of Insulation for motor	: Class F with temp. Rise limited to Class B
1.14.7 Motor RPM	: 1410
1.14.8 Pump design flow	: 1.8 LPM
1.14.9 Pump pressure	: 200 (max) Kg/cm²g
1.14.10 Safety valve set pressure	: 90 Kg/cm²g
1.14.11 in built safety valve	: Yes
1.14.12 Pump type	: Gear
1.14.13 Valves required	: By vendor
1.14.14 Volume of oil receiver	: 5 Lit
1.14.15 Dimensions of power pack Unit	: 400 H x 560 L x 340 W mm
1.14.16 First fill of oil / Qty	: By Vendor
1.14.17 Hydraulic tubing	: By Vendor
1.14.18 Instrumentation	: By vendor
1.14.19 Control panel	: By vendor
1.15 Empty weight	: 1100 Kg

INTRODUCTION: -

The hydraulic Dock Leveler is the result of many years' experience in materials handling engineering. "NIHVA" offers the best possible factor of safety with simple, easy to use controls. The high quality of manufacture gives a robust construction and many years of useful life. Each lip and platform movement can be individually controlled (or corrected) by push button.

The dock leveler can be set at any position. During loading, the Dock Leveler is in floating mode i.e. it follows the up/down movements of the vehicle deck. Even with lopsided vehicles, the Dock leveler lip will remain in contact, and toe-traps are avoided by the tilt built into the Dock Leveler.

The whole platform is designed to flex laterally adequately to adapt to sideways tilted vehicles. The leveler always seals perfectly to the lorry bed.

PRINCIPLE: -

Dock Leveler is an adjustable platform used between the transport vehicle and dispatch area of the warehouse of factory for efficient and safe material transfer applications. It bridges the gap between the dispatch area and the transport vehicle and compensates for variations in the truck bed height.

CONSTRUCTION: -

It basically consists of heavy duty fabricated Top Cheq. platform and bottom angle channel construction frame. Both are joined by stainless steel machined pins. Lip is hinged to the top platform by means of stainless-steel pins.

Top Platform is reinforced with I Beam welded below the platform to strengthen the platform. The cylinder is mounted at the best advantage point against the top platform and bottom frame. Compact power pack includes a single HP motor with overload protection. Hydraulic Pump is Dowty make. Hydraulic operations are by using solenoid control valve.

FEATURES: -

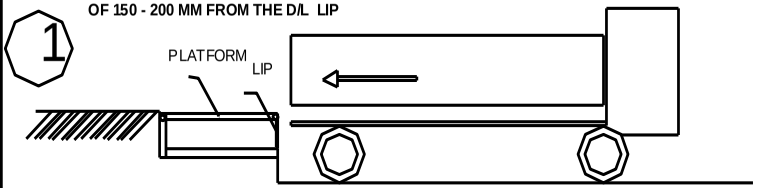
1. Convenient location of all controls.
2. Independent deck and lip.
3. Controls for quicker cycling time.
4. Smooth dependable hydraulics.
5. Quite operation at the touch of the button.
6. Years of trouble-free service.
7. Lower lift time cost and fewer maintenance requirements.
8. Recommended for all applications.
9. Maintenance Strut.
10. Hazard Stripes.
11. Toe guards.

PRE – OPERATING INSTRUCTIONS:-

1. Ensure Hydraulic Oil in the power pack.
2. Ensure Motor connection and proper earthing.
3. Ensure joint leakage.
4. Ensure proper location of platform and lip.

1

REVERSE THE TRUCK UPTO DISTANCE OF 150 - 200 MM FROM THE D/L LIP

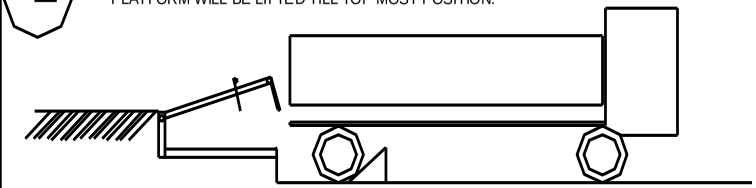


NOTE : ENSURE THAT EDGES OF TRUCK LOAD BODY ARE OUTSIDE LIP WIDTH I.E. CHECK ALIGNMENT OF LIP WITH TRUCK BODY.

2

HOLD PLATFORM UP BUTTON IN PRESSED POSITION.

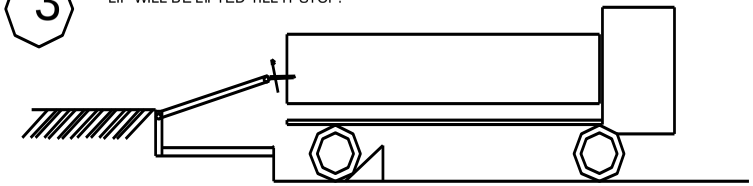
PLATFORM WILL BE LIFTED TILL TOP MOST POSITION.



3

HOLD LIP UP BUTTON IN PRESSED POSITION.

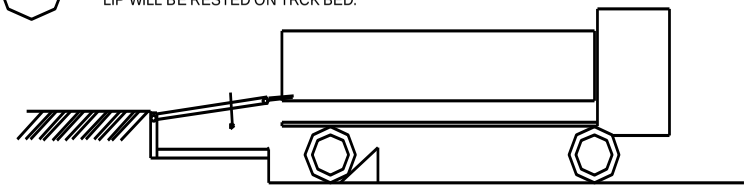
LIP WILL BE LIFTED TILL IT STOP.



4

HOLD PLATFORM DOWN BUTTON IN PRESSED POSITION.

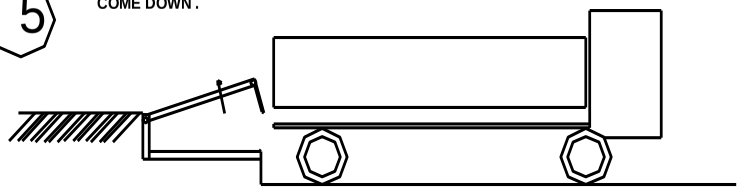
LIP WILL BE RESTED ON TRCK BED.



UNLOAD OR LOAD THE MATERIAL FROM THE TRUCK.

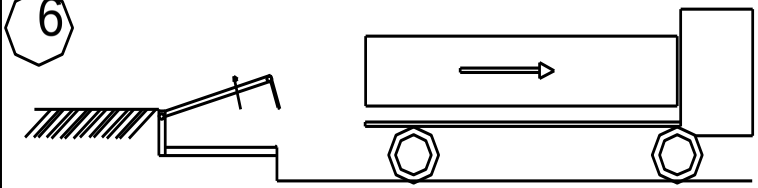
5

PRESS PLATFORM UP TO CLEAR THE LIP FROM TRUCK BED THAT TIME THE LIP WILL COME DOWN .



6

DRIVE THE TRUCK AWAY FROM D/L.



7

HOLD PLATFORM DOWN BUTTON .

LIP WILL BE PARKED TO ITS ORIGINAL POSITION.



OPERATING INSTRUCTION FOR DOCK- LEVELER.

ROUTINE MAINTENANCE AND LUBRICATION

These recommendations are applicable to the Dock Leveler considering average daily usage of 15 to 20 times. If it is used more frequently, the same time intervals for the maintenance should be adjusted accordingly.

DAILY: -

1. Check Hydraulic pipe lines for leak proof working.
2. Check electrical connections for their proper working.

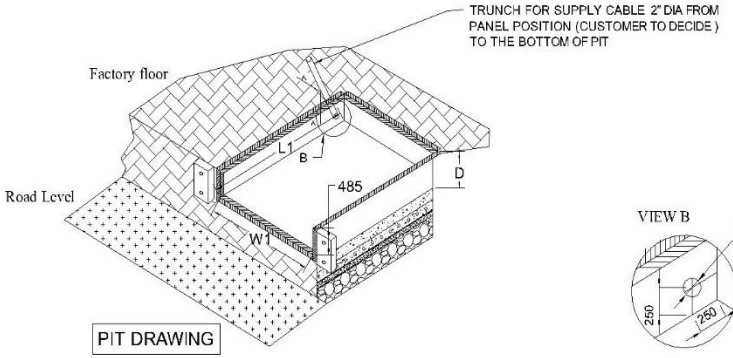
MONTHLY: -

1. Grease the hinge pins at bottom frame and top plate joint and extension lip plate hinge joint.
2. Grease the main Jack Cylinder and Lip Jack Cylinder, Front and Rear Trunion bearings.
3. Change Hydraulic Oil.

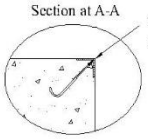
ASSEMBLIES
&
SPARE PARTS LIST

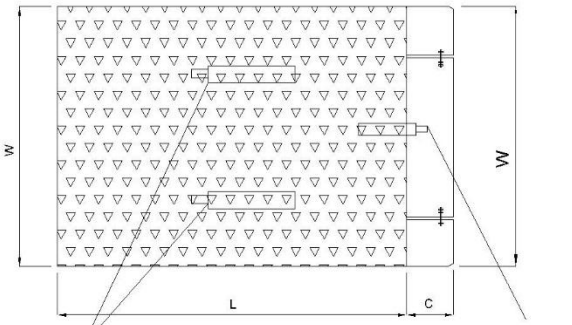
DOCK LEVELER SIZE	WIDTH (W)	LENGTH (L)	HEIGHT (H)	LIP (C) LENGTH	BUFFER SIZE (OPTIONAL)
	2070	2940	480	400	200X300X100

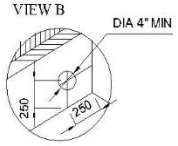
PIT DIMENSION	WIDTH (W 1)	LENGTH (L 1)	DEPTH (D)	ALL DIMENSIONS ARE IN mm
	2120	2990	485	

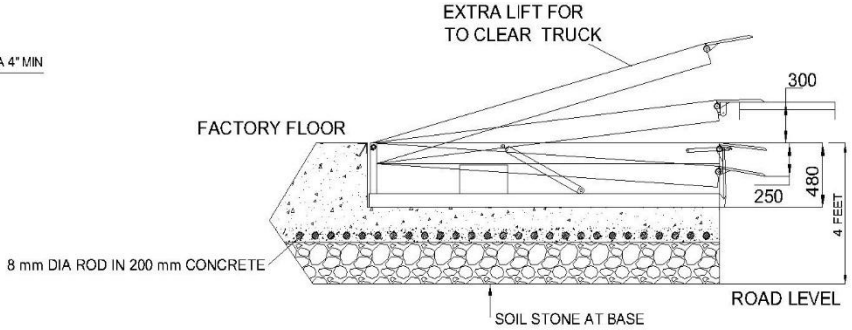


PIT DRAWING









REV NO	DATE	DESCRIPTION
REV -0	09/01/18	INITIAL DRAWING
REV -1	11/01/18	DOCK LEVELER SIZES REVISED AS PER ACTUAL PIT DIMENSIONS

DOCK LEVELER- 6 TON CAPACITY

NOTE: -

1. PROPER DRAINAGE SHOULD BE DONE
2. TRENCH AS SHOWN SHOULD BE MADE BY DECIDING THE POWER PACK POSITION (SUPPLY CABLE LENGTH STD. 7000 MM)
3. LEVEL OF THE PIT SHOULD BE PLAIN AND SHOULD TAKE AT LEAST 10000 KG FLOOR LOADING
4. MOUNTING ARRANGEMENT FOR BUFFERS SHOULD BE MADE AS PER BUFFER DRAWING (IF ORDERED)
5. FABRICATION TOLERANCE FOR PLATFROM SIZE WILL BE ± 15 MM

CLIENT APPORVAL-	DATE	SIGN.
DRAWN :- OMKAR	09/01/2018	
CHECKED :- S.S.	09/01/2018	
APPROVED :- S.S.	09/01/2018	
ALL DIMENSION ARE IN mm		
If any queries contact : design@nihva.com		

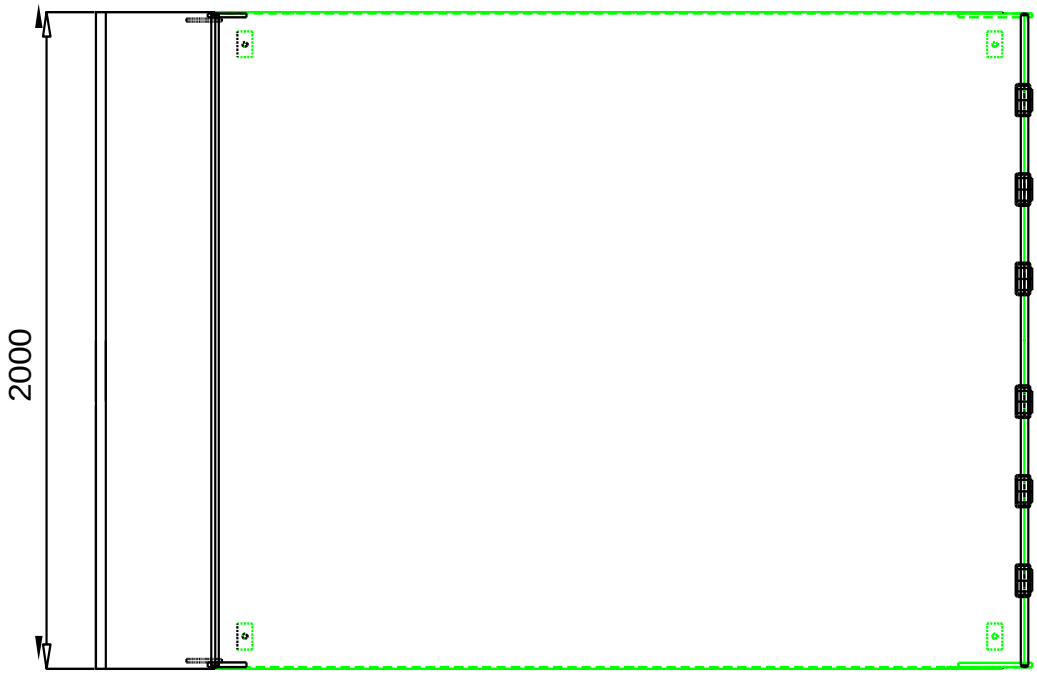
NIHVA TECHNOLOGIES PVT. LTD.
NIHVA TECHNOLOGIES PVT. LTD. PUNE
Tel : 020 66304033
Email : Info@nihva.com
Website :- www.nihva.com

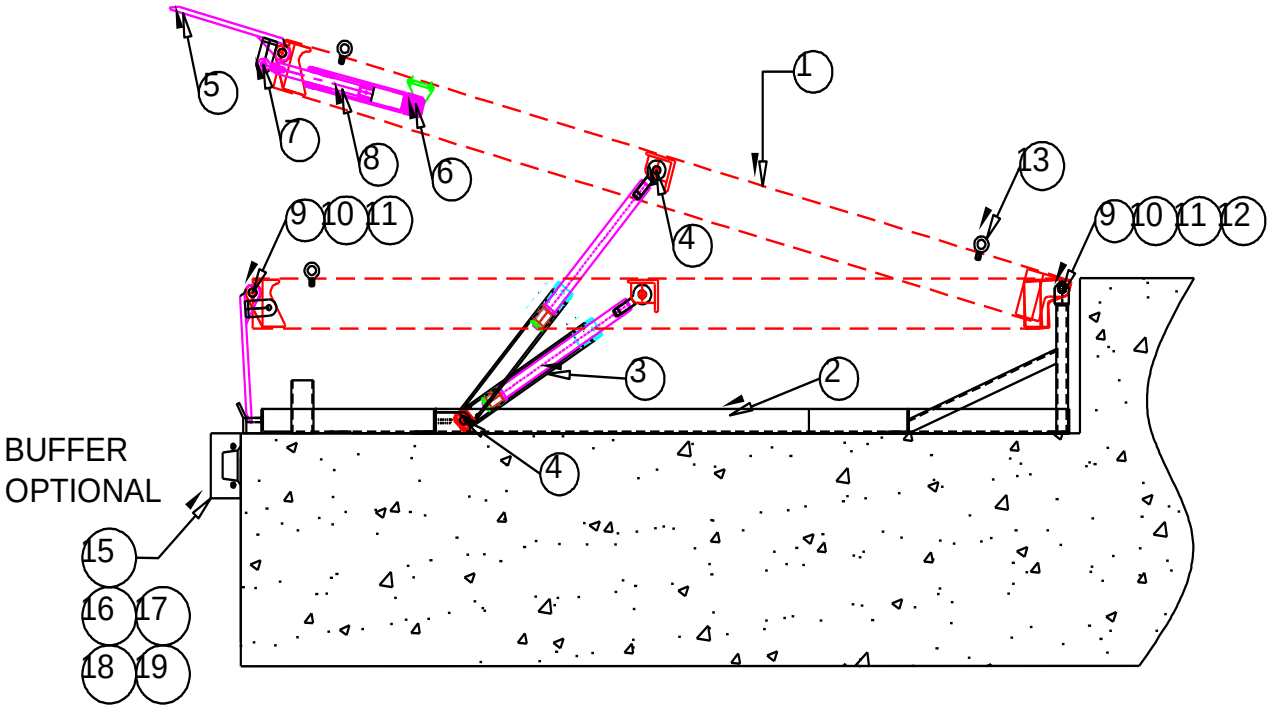
CLIENT - Zamil Construction India Pvt.Ltd.

PROJECT -

SITE -

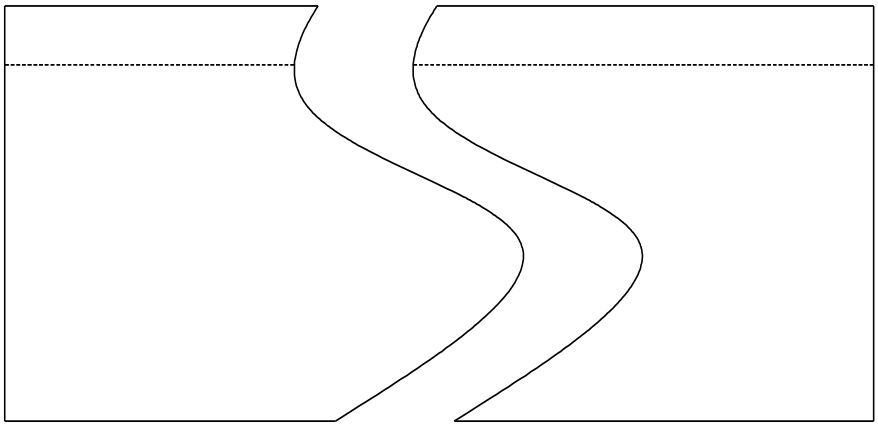
DWG. NO. - NIHVA/17-18/0331 | Rev- 1


2000


BUFFER
OPTIONAL
1
2
3
4
5
6
7
8
9
10
11
12
13
15
16
17
18
19

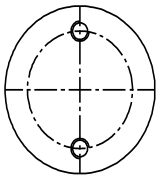
22	70000634	SIDE GUARD INNER (LH & RH)	1PAIR
21	70000633	SIDE GUARD MIDDLE(LH & RH)	1PAIR
20	70000632	SIDE GUARD OUTER(LH & RH)	1PAIR
19	10000309	ANCHOR FASTNER	8
18	70000631	RUBBER BUFFER MTG. ANGLE	4
17	PSL00001	SPLIT PIN	8
16	70000630	RUBBER BUFFER PIVOT PIN	4
15	70000629	RUBBER BUFFER	2
14	10000309	ANCHOR FASTNER	8
13	10000137	I BOLT M16	4
12	NGR00001	GREASE NIPPLE	6
11		M6x20L CSK BOLT	8
10	70000164	PIN LOCKING WASHER	2
09	70001211	PLATFORM & LIP PIVOT PIN	2
08	70000383	LIP JACK ASSLY.	1
07	70000167	LIP JACK TOP PIVOT PIN	1
06	70000168	LIP JACK BOTTOM PIVOT PIN	1
05	70001212	LIP PLATE	1
04	70000165	MAIN JACK PIVOT PIN	4
03	70000156	MAIN JACK ASSLY.	2
02	70000627	BOTTOM FRAME WELDED ASSLY	1
01	70000626	TOP PLATFORM WELDED ASSLY	1
SR. NO	PART No.	DESCRIPTION	QTY.

DETAILS:-5
LIP PLATE
PART NO-70001212
QTY-01

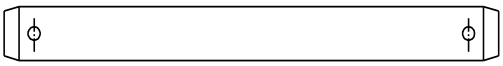




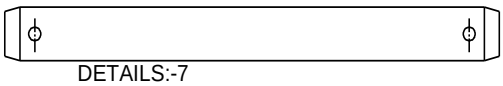
DETAILS:-9
PLATFORM & LIP PIVOT PIN
PART NO-70001211
QTY-02



DETAILS:-10
PIN LOCKING WASHER
PART NO-70000164
QTY-04



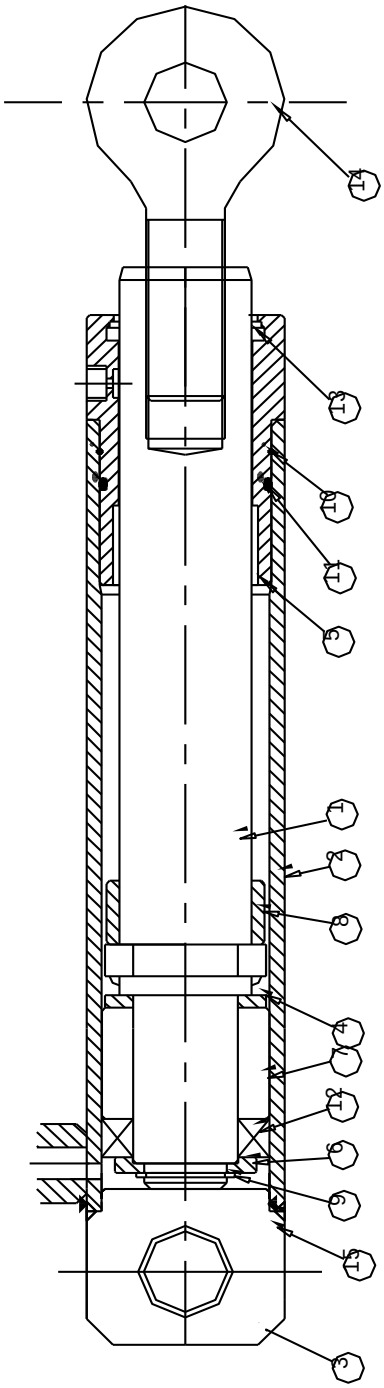
DETAILS:-6
LIP JACK PIVOT PIN(BOTTOM)
PART NO-70000168
QTY-01



DETAILS:-7
LIP JACK PIVOT PIN(TOP)
PART NO-70000167
QTY-01



DETAILS:-4
MAIN JACK PIVOT PIN
PART NO-70000165
QTY-02

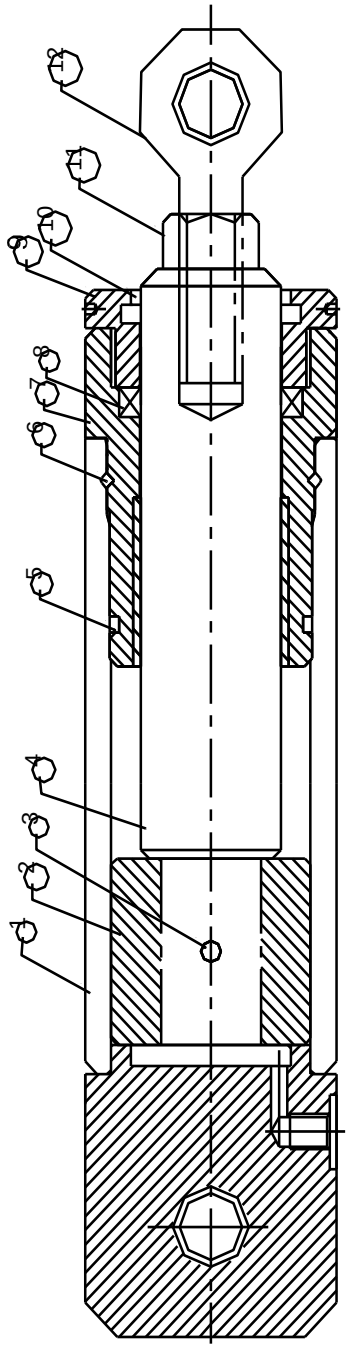


SR.NO.	ITEM NO.	PARTS NO.	DESCRIPTION	SIZE	QTY
15	15	10000071	SINTERED BUSH	1"X1.1/8"X1"	2
14	14	10000070	ROD END BEARING	M24X2X25PIN	1
13	13	WPS00015	WIPER SEAL	Ø40xØ48x7	1
12	12	PTS00011	PISTON U SEAL	Ø50xØ40x9	1
11	11	ORN00040	O RING	Ø48X2.5	1
10	10	LKW13225	LOCKING WIRE	13 GAUGE	1
9	9	ECP00009	EXTERNAL CIRCLIP	EXT Ø25	1
8	8	70000050	SPACER Ø40	Ø40XØ49XSP	1
7	7	70000154	PISTON Ø40	Ø50.8XØ31.75X50	1
6	6	70000049	WAHSER RETAINER SEAL	Ø48XØ25.2X5	1
5	5	70000048	WIPER SEAL HOUSING	Ø63X85 L	1
4	4	70000047	WASHER	Ø49XØ32X4	1
3	3	70000159	BOTTOM END	Ø63X49	1
2	2	70000158	ASSLY WELDED CYLINDER	2" HONED	1
1	1	70000157	SHAFT JACK	Ø40 x 580	1

DETAILS:-3

MAIN JACK ASSLY

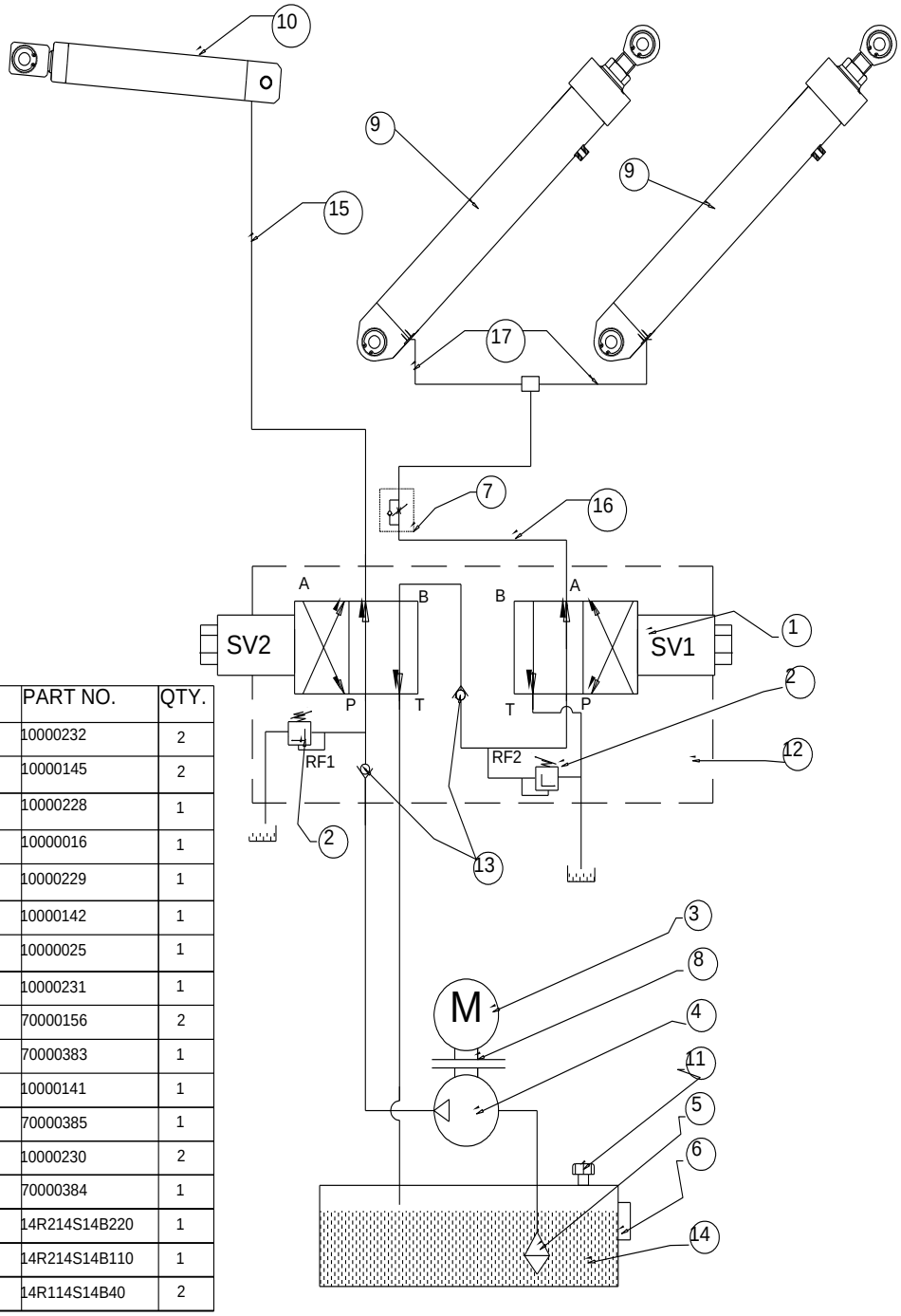
PART NO-70000156



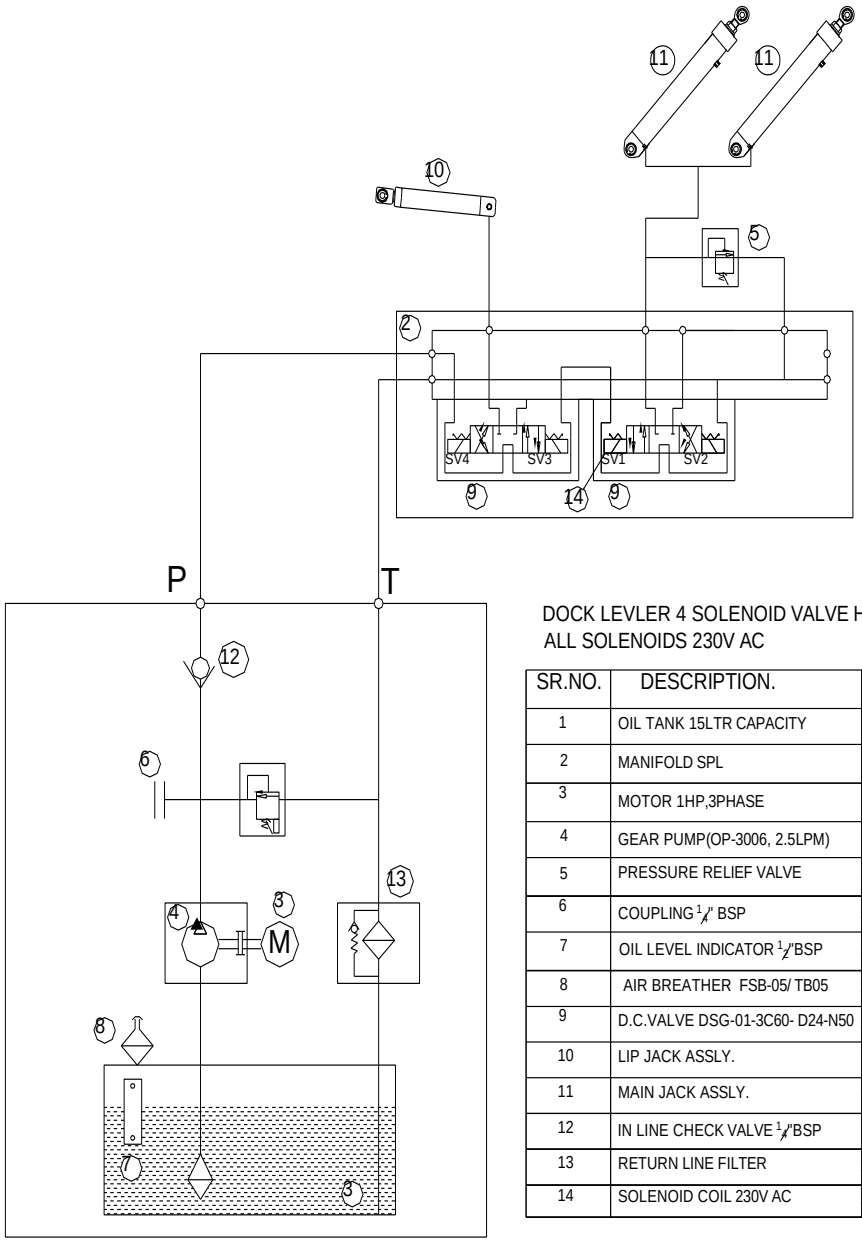
SR.NO.	PARTS NO.	DESCRIPTION	QTY
12	REB00006	ROD END BEARING	1
11		HEX NUT M-16X2P	1
10	WPS00003	SEAL WIPER	1
9	70000390	WIPER SEAL HOUSING	1
8	RDS00003	SEAL ROD	1
7	70000389	SEAL HOUSING	1
6	LKW13235	WIRE LOCKING	1
5	ORN00002	O RING	1
4	70000388	SHAFT(PLATED)	1
3	DSP00001	SPRING PIN	1
2	70000387	PISTON	1
1	70000386	JACK CYLINDER	1

DETAILS:-8
LIP JACK ASSLY
PART NO-70000383

HYDRAULIC CIRCUIT
FOR DOCK LEVELER



SR.NO.	DESCRIPTION.	PART NO.	QTY.
1	SOLENOID VALVE (SV1,SV2)	10000232	2
2	RELIEF VALVE (RF1,RF2)	10000145	2
3	MOTOR 1HP,3PHASE	10000228	1
4	GEAR PUMP(OP-3006)	10000016	1
5	SUCTION STRANER	10000229	1
6	OIL LEVEL INDICATOR	10000142	1
7	THROTTLE VALVE	10000025	1
8	LOVEJOY COUPLING	10000231	1
9	MAIN JACK ASSLY.	70000156	2
10	LIP JACK ASSLY.	70000383	1
11	BREATHER FSB-25	10000141	1
12	MANIFOLD BLOCK	70000385	1
13	IN LINE CHECK VALVE $\frac{1}{4}$"	10000230	2
14	OIL TANK 15LTR CAPACITY	70000384	1
15	BLOCK TO LIP JACK HOSE	14R214S14B220	1
16	BLOCK TO MAIN JACK HOSE	14R214S14B110	1
17	T BLOCK TO MAIN JACK HOSE	14R114S14B40	2



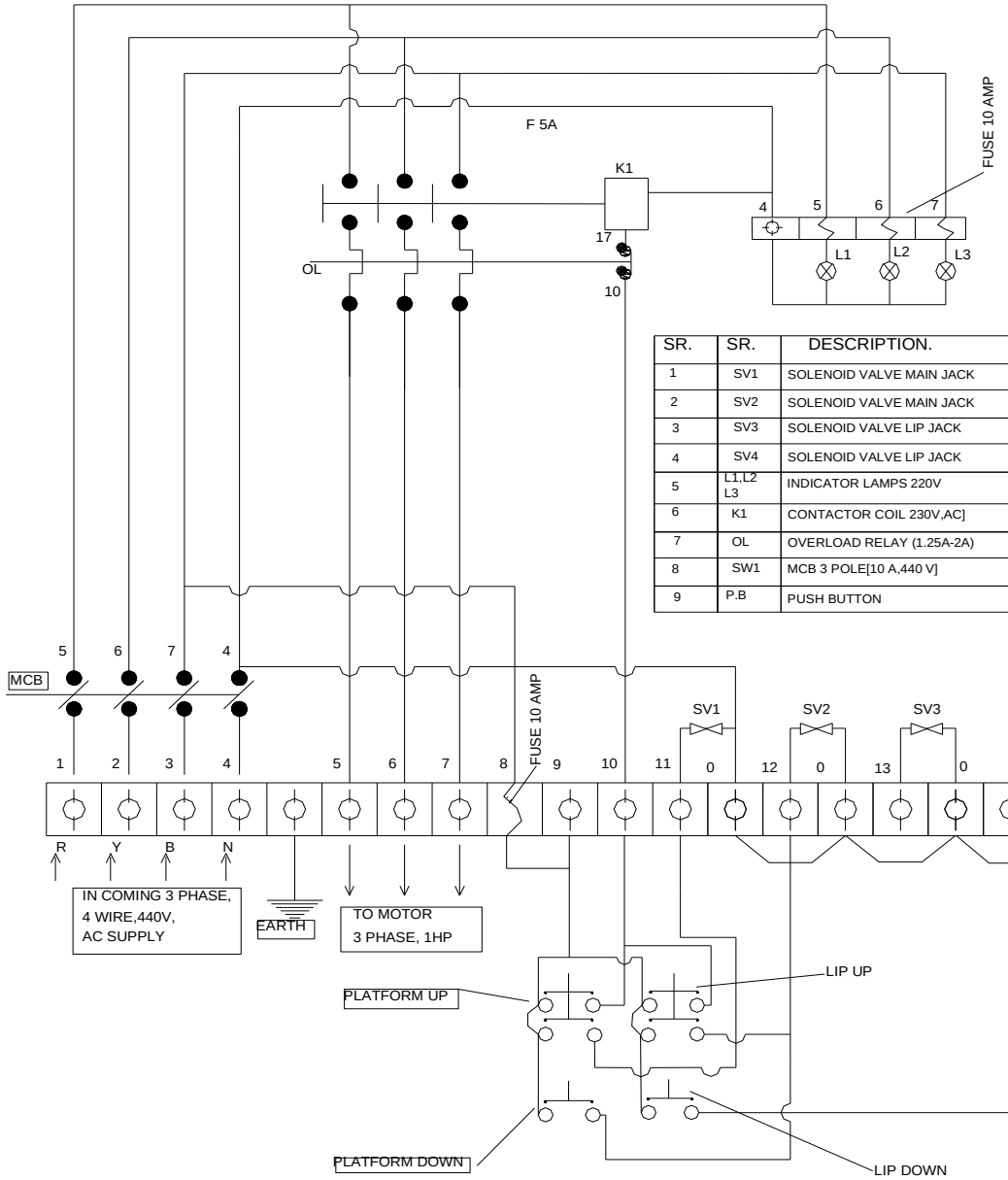
DOCK LEVLER 4 SOLENOID VALVE HYD.CIRCUIT
ALL SOLENOIDS 230V AC

SR.NO.	DESCRIPTION.	PART NO.	QTY.	MAKE
1	OIL TANK 15LTR CAPACITY	70000384	1	TEPL
2	MANIFOLD SPL	70000385	1	TEPL
3	MOTOR 1HP,3PHASE	10000228	1	CG OR BB
4	GEAR PUMP(OP-3006, 2.5LPM)	10000016	1	DOWTY
5	PRESSURE RELIEF VALVE	10000145	2	POLYHYDRON
6	COUPLING $\frac{1}{4}$" BSP	10000142	1	ZEC
7	OIL LEVEL INDICATOR $\frac{1}{2}$"BSP	10000142	1	TEPL
8	AIR BREATHER FSB-05/ TB05	10000141	1	HYDROLINE
9	D.C.VALVE DSG-01-3C60- D24-N50	10000232	2	YUKEN
10	LIP JACK ASSLY.	70000383	1	TEPL
11	MAIN JACK ASSLY.	70000156	2	10000141
12	IN LINE CHECK VALVE $\frac{1}{4}$"BSP	70000385	1	HYDROLINE
13	RETURN LINE FILTER	10000258	1	HYDROLINE
14	SOLENOID COIL 230V AC	10000793	1	HYDROLINE

DOCK LEVLER PUSH BUTTON OPERATION

SR. NO.	PUSH BUTTON OPERATION	SOLENOID VALVE ACTUATION	MOTOR CONTACTOR	FUNCTION
1	PUSH BUTTON PLATFORM UP	SOLENOID VALVE 2 (SV 2)	MOTOR COLCKWISE	OPENS MAIN JACK (PORT OPEN TO PUMP ONE)
2	PUSH BUTTON PLATFORM DOWN	SOLENOID VALVE 1 (SV 1)	MOTOR OFF	LOWERS MAIN JACK (PORT OPEN TO TANK)
3	PUSH BUTTON LIP UP	SOLENOID VALVE 3 (SV 3)	MOTOR COLCKWISE	OPENS LIP JACK (PORT OPEN TO PUMP LINE)
4	PUSH BUTTON LIP DOWN	SOLENOID VALVE 4 (SV 4)	MOTOR OFF	LOWERS LIP JACK (PORT OPEN TO TANK)

DOCK LEVLER 4 SOLENOID VALVE AC PANEL



SR.	SR.	DESCRIPTION.	PART NO.	QTY.
1	SV1	SOLENOID VALVE MAIN JACK	10000336	1
2	SV2	SOLENOID VALVE MAIN JACK	10000336	1
3	SV3	SOLENOID VALVE LIP JACK	10000336	1
4	SV4	SOLENOID VALVE LIP JACK	10000336	1
5	L1,L2 L3	INDICATOR LAMPS 220V	10000337	3
6	K1	CONTACTOR COIL 230V,AC]	10000338	1
7	OL	OVERLOAD RELAY (1.25A-2A)	10000339	1
8	SW1	MCB 3 POLE[10 A,440 V]	10000340	1
9	P.B	PUSH BUTTON	10000024	1

SR. NO.	PUSH BUTTON OPRATING TABLE	I/P	O/P-1	O/P-2
1	PUSH BUTTON PLATFORM UP	COMMAN WIRE NO.09	MOTAR CONTACTOR TO WIRE NO. 10	S/V 1 WIRE NO. 11
2	PUSH BUTTON PLATFORM DOWN	COMMAN WIRE NO.09	NA	S/V 2 WIRE NO. 13
3	PUSH BUTTON LIP UP	COMMAN WIRE NO.09	MOTAR CONTACTOR TO WIRE NO. 10	S/V 3 WIRE NO. 15
4	PUSH BUTTON LIP DOWN	COMMAN WIRE NO.09	NA	S/V 4 WIRE NO. 17

NEW MODIFY DOCK LEVELER ELECTRIC CIRCUIT:-

